

Derwent Patent Search — Summary of Prior Art Analysis

1. Software Search — Closest Prior Art

US12094611B2 — "Deep Learning Based Fetal Heart Rate Analytics" (GE Precision Healthcare, granted 2024)

- Strongest prior art found overall — the only granted (not just filed) patent among the close matches.
- Trains on CTG tracings, then uses longitudinal correlation between patterns at different time points to confirm events in real time.
- Outputs a derived set of parameter values — a step toward parameter-level output, not just a single label.
- Does NOT use multi-task branches, FIGO/NICE guideline infusion, or SHAP/IG interpretability.

IN202641063292A — Hybrid Deep Learning Fetal Monitoring & Preterm Labor Prediction System

- Most architecturally complex filing reviewed: WSGF filtering → RGO-ICA source separation → DMM segmentation → GAN-built 3D correlation graph → CARANet classifier → LSTM preterm prediction.
- Classifies into 5 distress sub-classes — closer to parameter-level granularity, but still one classification head, not independent per-parameter branches.
- No guideline infusion, no SHAP/IG interpretability.

CN117017216B — Fetal Distress Diagnosis via Multi-Mode Feature Fusion

- Closest analog to "knowledge infusion" across all three searches: fuses an expert-feature branch with a learned signal branch (SE-TCN) via cross-modal attention.
- "Expert knowledge" is generic morphological/statistical features, not codified FIGO/NICE criteria.
- Output is a single diagnosis, not decomposed clinical parameters.

2. Hardware Search — Closest Prior Art

- US7818050B2 (Lono Medical, granted) & US5524631A (NASA, 1994): passive acoustic FHR sensing with basic filtering — no ICA source separation, no UC sensing, no ML.
- IN201104480I4 (GE): Doppler-based handheld device, no signal separation or ML — mainly signals GE's broad incumbent footprint in this space.

3. Combination Search — Closest Prior Art

- IN202641050189A (RNS Institute, filed April 2026): closest overall match — wearable sensor-fusion patch with on-device ML risk prediction. But uses PPG-referenced template subtraction (not ICA/BSS), no UC sensing channel, single risk score only.
- CN120323948A (transformer-based FHR+UC cross-attention fusion) and IN202441061552A (wearable fPCG + onboard AI classifier): each close on one architectural front, but single-task with no guideline infusion or parameter decomposition.

Summary of Patent search analysis

- No single record combines: multi-task clinical-parameter decomposition + FIGO/NICE guideline infusion + temporal early-warning prediction + dual-path UC sensing.
- Strongest individual risk: US12094611B2 (GE, granted).
- Most time-sensitive to monitor: IN202641050189A (RNS Institute).